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MyDesign Portfolio

Element B Initial Template

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Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Wet Pipe Fire Sprinkler System>

Source

National Fire Protection Association (NFPA). NFPA 13: Standard for the Installation of Sprinkler Systems.

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some [free online journals and research databases](#)), professional organizations, government agencies, local and national newspapers, patents (here is a [patent search site for the United States Patent and Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image

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Include at least one image of this prior design attempt. Make sure to cite all images.

Chemical Name: _____	
HEALTH HAZARD 4. Deadly 3. Extreme Danger 2. Hazardous 1. Slightly Hazardous 0. Normal Material	FIRE HAZARD (FLASH POINT) 4. Below 73°F 3. Below 100°F 2. Below 200°F 1. Above 200°F 0. Will Not Burn
USE NO WATER IN Oxidizer Simple Asphyxiant Acid Alkali Corrosive Radiation Hazard	REQUIRED PERSONAL PROTECTIVE EQUIPMENT
SPILLING HAZARD 4. May Detonate 3. Shock And Heat May Detonate 2. Violent Chemical Change 1. Unstable If Heated 0. Stable	☐ Safety Glasses ☐ Splash Goggles ☐ Face Shield & Eye Protection ☐ Dust Respirator ☐ Vapor Respirator ☐ Gloves ☐ Synthetic Apron ☐ Full Suit ☐ Boots ☐ Other: _____

Details

A wet pipe sprinkler system is a common fire suppression system used in commercial and residential buildings. Water is always stored in the pipes. When heat from a fire reaches a certain temperature (usually 155°F/68°C), the glass bulb in the sprinkler head bursts, releasing water directly onto the fire.

- Detects: Heat-sensitive glass bulb
- Alerts: Connected fire alarm system activates building alarms
- Suppresses: Sprays water directly onto flames

Analysis

Even though they are effective, wet pipe systems have limits as they may cause water damage to property even in small fires. And they are not suitable for freezing environments, as water in pipes can freeze. And they only activate when heat reaches the sprinkler head, meaning early smoke detection may not occur. Also, the system does not always provide early detection before flames grow.

Prior Design Attempt #2 - <Smoke Detector with Integrated Alarm>

Source

U.S. Consumer Product Safety Commission (CPSC). Fire Safety Guidelines.

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Image*Details*

Smoke detectors sense the smoke particles in the air using ionization or photoelectric sensors. When the smoke reaches a certain concentration, the device sounds a loud alarm to alert the people.

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- Detects: Smoke particles
- Alerts: Loud audible alarm
- Suppresses: They don't suppress fire

Analysis

Smoke detectors are very useful for early warning but have major limits as:

- They do not suppress fires.
- They can produce false alarms from cooking smoke.
- They depend on people responding quickly.

This system only halfway solves the problem (detection and alert), but it doesn't extinguish fires.

Prior Design Attempt #3 - <Fire Alarm Control Panel with Sprinkler Integration>

Source

Honeywell International – Commercial Fire Alarm Systems Documentation.

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Image

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Details

Modern fire alarm control panels connect smoke detectors, heat detectors, and sprinkler systems into one integrated network. They detect smoke and heat sensors. They can alert to audible alarms, flashing strobes, and emergency notifications. And they can suppress the activation of sprinkler systems. And they're mainly used in schools, hospitals, and office buildings.

Analysis

Although they can be good, they have some weaknesses, including that they have a high installation cost and require professional maintenance. And they are not always accessible for small residential homes. While they can be effective, affordability and maintenance challenges might limit widespread use.

Prior Design Attempt #4 - <Automatic Fire Suppression Ball (Elide Fire Ball)>

Source

Elide Fire – Product Specifications.

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some [free online journals and research databases](#)), professional organizations, government agencies, local and national newspapers, patents (here is a [patent search site for the United States Patent and](#)



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[Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image



Details

The Elide Fire Ball is a small ball-shaped device filled with extinguishing powder. When it comes into contact with flames for a few seconds, it automatically explodes and releases fire-suppressing chemicals. It can detect direct contact with flame and can alert you by making a loud explosion sound. It can also suppress the fire by releasing extinguishing powder.

Analysis

While I think it's a really creative idea, it can also be one of the limiting factors as it must be placed very close to the fire source, and it may not activate during slow-smoldering fires. It's also limited in the coverage area that it can put out the fire so it might not be that good. And mainly, it may not provide early detection or wide-area protection, which is bad as it could affect all life around and can make the fire spread quickly if it can't notify someone before it spreads.

Prior Design Attempt #5 - <Fire Extinguisher (ABC Type)

Source

Occupational Safety and Health Administration (OSHA). *Portable Fire Extinguisher Standards* (29 CFR 1910.157).

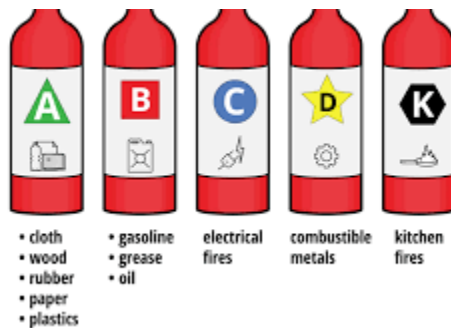
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Image



Details

ABC fire extinguishers are portable extinguishers that discharge dry chemical powder to suppress Class A (ordinary combustibles), B (flammable liquids), and C (electrical) fires. It can't detect automatically, and there is no automatic alert. And while yes, it can suppress, it would happen if it were manually used, which is probably as it might be able to be accessed in time, which could hurt a lot of people.

Analysis

This design doesn't really meet the full requirements for automatic detection, alerting, and suppression. Although it's effective when used properly, fire extinguishers rely entirely on people alone. It has no automatic detection, no automatic alert system, and it can even require someone who has been trained and present to even be at a fire, which may definitely not be the case all the time.